

methods often face a trade-off between task complexity and analytical specificity (Wang et al., 2022; Bricken et al., 2023; Bereska and Gavves, 2024).

Approach: A "Divide-and-Reconstruct" Framework. We propose a multi-level framework to enable scalable and interpretable analysis of LLM numerical reasoning. The core idea is to connect high-level task performance with its low-level building blocks. Our long-term vision is to use this framework to identify persistent reasoning traits (e.g., recurring error patterns) and trace their causal roots across layers of analysis: from observable behavior (task success/failure, accuracy), to self-explanation (the model’s rationale), and finally to internal computation (model internals). Inspired by cognitive science and mechanistic interpretability, we adopt a “divide-and-reconstruct” strategy: first, we decompose a complex task into its elementary skills, and then we evaluate LLMs on both the isolated skills and their integration.

Application: Probing Numerical Reasoning. As an application of this framework, we investigate the limits of LLM numerical reasoning. We focus on the known challenge of problems requiring heuristic search (Yao et al., 2023). Following our "divide-and-reconstruct" strategy, we first test LLMs on elementary mathematical operations. We then examine their ability to integrate these skills in two progressively complex tasks: (a) primality checking, which can be solved by a deterministic search algorithm (i.e., trial division) with a variable number of steps, and (b) the Game of 24, which requires a non-deterministic, combinatorial search. This progression allows us to trace how foundational numerical skills (or their fragility) contribute to success or failure on more complex problems.

4 Experimental Methods

4.1 Problem Types

Our test contained 100 problems divided equally into four sets by problem type.

Set 1: Basic Mathematical Operations This set allows us to set a reference for LLM mathematical skills in problems that involve only four basic mathematical operations – addition, subtraction, multiplication, and division – and require minimal reasoning. We begin with simple problems (e.g., $2 + 2 + 4 - 2 = ?$ and $1/13 = ?$) before increasing difficulty by involving longer chains of operations. Some problems involve transcendental numbers (e.g., π and e) and reciprocals of primes to exploit a numerical precision problem observed in LLMs.

Set 2: Advanced Mathematical Operations We introduce more advanced operations and concepts, such as exponentiation, logarithms, and complex numbers. Most problems require combinations of operations. Some problems contain slightly unusual values, such as fractions as exponents (e.g., $400^{0.23}$) and logarithms with base 5 or 13 (e.g., $\log_{13} 169$). This set provides a sterner test of LLM abilities in executing chains of operations than Set 1, but still offers a minimal scope of reasoning.

Set 3: Check if a Number is Prime We ask the agents to check the primality of numbers. We varied the length of the numbers (discontinuously from 1- to 54-digit numbers). Some numbers are relatively famous and likely to be discussed in LLM training data, such as Mersenne Primes of the form $2^p - 1$ (e.g., $2^{13} - 1$ and $2^{31} - 1$). We also include some non-primes of the same form ($2^{27} - 1$ and $2^{29} - 1$), which we refer to as the *Mersenne Prime Lookalikes*.

This set allows us to examine LLM performance in combining basic skills to solve a more complex problem. The algorithms for checking the primality of numbers by searching for factors are well-known; however, the point at which to stop searching is non-deterministic. There are explicit clues in (prime) number theory to exploit – such as any number of the form $2^p - 1$ cannot be prime if p is not prime (i.e., $2^{27} - 1$ is not prime) – or more implicit ones in our number sense (e.g., we can tell 333 or 77777 are not prime without explicit reasoning).

Set 4: The 24 Game 24 is a popular number game that has been previously used in teaching children and testing LLMs (Van der Maas and Nyamsuren, 2017; Ding et al., 2023). The game requires only elementary mathematical knowledge to play. A player is given a set of four numbers (e.g., $[2, 4, 6, 6]$) and their goal is to produce 24 as the output, using each number once, and the four basic operations – addition, subtraction, multiplication, and division – as many times needed (e.g., $(6 + 2 - 4) \times 6 = 24$).

Despite its simplicity, the search space is vast with numerous combinations of numbers and operations to consider. This space can be reduced using number relationships as clues. For example, a helpful strategy is to look for factors that make up 24 or for multiples of 24. In the above example, we may

Table 1: Five Models included in the **main study** and their versions. These models were tested over a 7-day period between December 30, 2024 – January 5, 2025 PST.

MODEL	VERSION	REASONING ENABLED?
OPENAI CHATGPT	O1-MINI	YES
OPENAI CHATGPT	O1	YES
GOOGLE GEMINI	1.5	NO
ANTHROPIC CLAUDE	SONNET 3.7	NO
MICROSOFT COPILOT	-	NO

set aside one of the 6s and try to manipulate the rest of the numbers to get the required $24/6 = 4$. There are numerous such strategies, but none of them guarantee success; rather, solvers need to find the solutions through an informed trial-and-error search using their “number sense.”

4.2 Participants

We began with eight LLMs and ran an initial test of competence on basic mathematical operations for inclusion in the main test. Three models – ChatGPT 4o, Cohere AI, and Mistral AI – were excluded as they scored below our threshold of 80% accuracy, and the remaining five (Table 1) proceeded to the main study. To validate our findings, we later ran a follow-up test with three models (Table 2).

Notably, the agents differ substantially in architecture, and the differences further propagate due to fine-tuning for specific abilities. For example, both o1 models are equipped with enhanced reasoning, but the o1 model “thinks” longer using more chains than the o1-mini. The remaining three models were trained for general-purpose usage. Complete details are unavailable due to the proprietary nature of the models. Some high-level similarities are: (1) all use transformer models pre-trained on large corpora of texts, and (2) follow probabilistic principles for token generation using contextual information captured by positional encodings, self-attention mechanisms, and multi-head attention.

4.3 Experimental Procedure and Scoring

In the main study, we presented the agents with six prompts in sequence (Appendix A). First, we explained that they would be provided with some problem sets to solve and asked to show their steps before summarizing answers in ordered lists. In prompts 2-4, we presented the first three problem sets with brief task explanations. In prompt 5, we asked the agents about the 24 game and its standard rules. After confirming they had the intended version in mind, we gave them 25 games to play.

To examine the effects of complexity on reasoning in the 24 game, we conducted a follow-up study, creating a second, harder set of 24 games and contrasting LLM performance from the easier set used earlier. Here, we used two prompts in sequence (Appendix B). In the first prompt, we asked the LLMs to state the game rules and provided the same set of 25 games as in the main study (Set 4). In the second prompt, we provided 25 additional games of higher difficulty. As a notable difference from the first study, we explicitly instructed in the prompts not to use any coding tools to prevent solutions through exhaustive search, so that their ability to solve the problems efficiently through reasoning could be examined. The prompts proved sufficient to prevent the Gemini and Deepseek models from coding, but we needed to disable the coding abilities for the ChatGPT model.

Table 2: Three Models included in the **follow-up study** and their versions. These models were tested over a 5-day period between June 16, 2025 – June 20, 2025 PST.

MODEL	VERSION	REASONING ENABLED?
OPENAI CHATGPT	O3	YES
GOOGLE GEMINI	2.5 PRO	YES
DEEPSEEK	R1 (DEEPTHINK)	YES

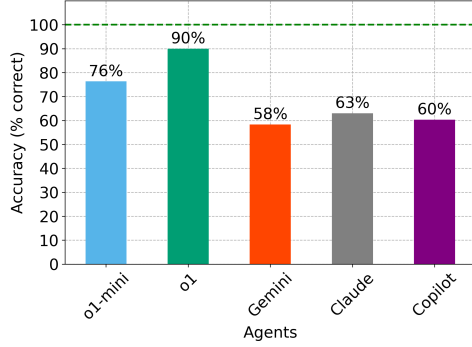


Figure 1: Overall performance of the models represented by the average score across problem sets.

We scored each correct answer by 1 point and a wrong answer by 0. For decimal fractions as answers, we asked for three digits after the decimal point, but we awarded the point if the first two digits were correct. Each agent performed the test three times, and the scores were averaged over three trials.

5 Results

5.1 Overall Test Performance of LLMs

The overall performance of the agents is shown in Figure 1. ChatGPT o1 model performed the best (90% average accuracy), followed by o1-mini (76%). The remaining three models performed at a level similar to each other (58-63%) but considerably poorer than the two ChatGPT models. The performance discrepancy is not surprising since the ChatGPT models are equipped with explicit reasoning abilities. However, this high-level view obscures their competence in specific mathematical tasks. Consequently, we cannot determine which 10% of the problems the o1 agent made mistakes in, or whether Claude, Gemini, and Copilot made mistakes in the same problems, despite their similar scores. In the following two sections, we focus on specifying the scope of their mistakes.

5.2 LLM Performance in Different Problem Types

Figure 2 shows LLM performance divided across problem types. All agents performed well on the basic operations (Set 1, Fig. 2a), with the lowest accuracy being 86%, showing all agents included in the test possess the elementary mathematical skills needed to perform the other tasks in the test. Therefore, the scores in the other tasks reflect how well they were able to generalize the elementary skills needed to solve more complex problems.

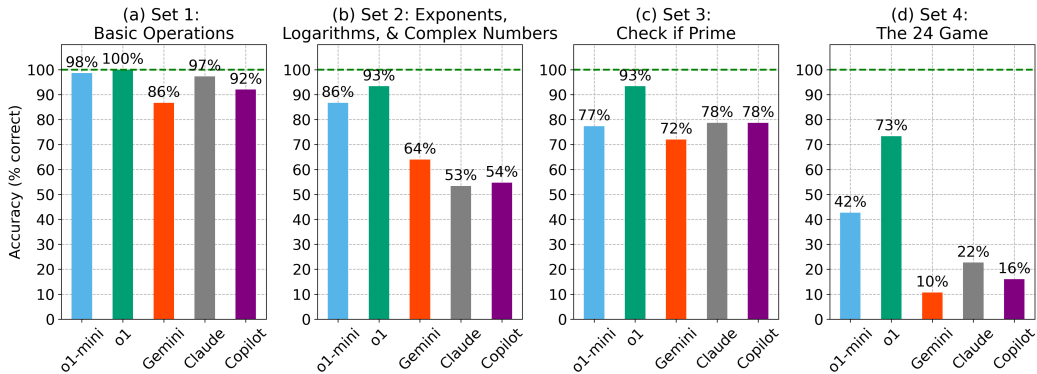


Figure 2: Performance of each agent in each of the four problem categories. The agents performed reasonably well in the first three sets but struggled in the fourth set, which contained the games of 24.